

LAB2: IoT Webserver with LED, Sensors, LCD, and Servo Control

1. Overview

In this lab, students will design an ESP32-based IoT system with MicroPython that integrates a web interface, sensors, an LCD display, and a servo motor. The system will allow users to control an LED and servo, read sensors, and send custom messages to the LCD through a webserver.

This lab emphasizes interaction between a web user interface and physical hardware, giving students practice in event-driven IoT design.

2. Learning Outcomes (CLO Alignment)

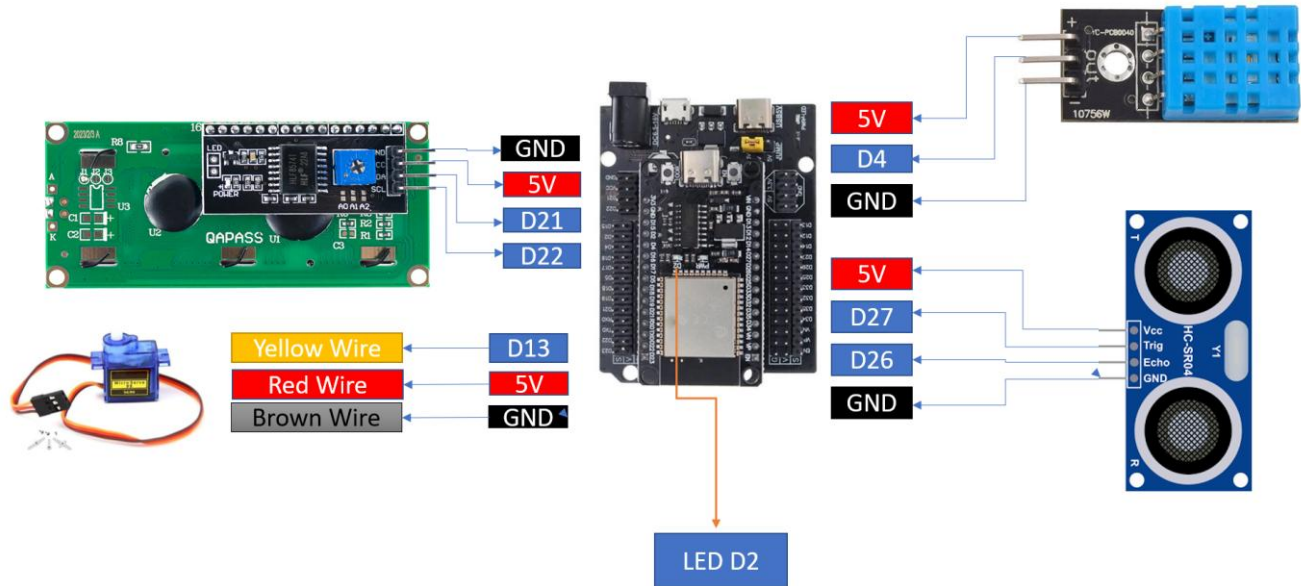
By the end of this lab, students will be able to:

- Implement a MicroPython webserver to serve HTML controls.
- Control an LED from the web page.
- Read data from DHT11 and ultrasonic sensors and expose it on the webserver.
- Use web buttons to selectively show temperature and distance on an LCD (I2C).
- Control a servo motor through a web slider and display its angle on the LCD.
- Send custom text from a textbox to display on the LCD.
- Document wiring, interface behavior, and system operation.

3. Equipment

- ESP32 Dev Board (MicroPython firmware flashed)
- DHT11 sensor (temperature/humidity)
- HC-SR04 ultrasonic distance sensor
- LCD 16x2 with I2C backpack
- SG90 servo motor
- Breadboard and jumper wires
- USB cable and laptop with Thonny
- Wi-Fi access

4. Wiring



5. Tasks and Checkpoints

Task 1 - Sensor Monitoring (15 pts)

- Read temperature and humidity from the DHT11 sensor.
- Measure distance using the HC-SR04 ultrasonic sensor.
- Display all sensor values on the web page and refresh them every 1-2 seconds.
- **Evidence:** Screenshot showing the sensor readings on the web page.

Task 2 - Sensor Data to LCD (15 pts)

- Add two buttons to the web page:
 - Show Distance: Display the ultrasonic distance on LCD line 1.
 - Show Temperature: Display the temperature on LCD line 2.
- **Evidence:** Photo showing the correct sensor values on the LCD.

Task 3 - Web-Controlled Servo (15 pts)

- Connect an SG90 servo motor to the ESP32.
- Add a slider to the web page with a range from 0 to 180 degrees.
- Moving the slider must change the servo angle.
- Display the selected angle on the web page.
- **Evidence:** Short video showing the web slider, servo movement.

Task 4 - Custom Text to LCD (15 pts)

- Add a textbox and a Send button to the web page.

- Text entered by the user must appear on the LCD.
- Text longer than 16 characters should scroll across the LCD.
- **Evidence:** Short video showing text sent from the browser to the LCD.

Task 5 - Complete IoT Web Dashboard Integration (20 pts)

- Combine all previous webserver tasks into one ESP32 program and one webpage.

Task 6 - Documentation and Demonstration (20 pts)

- Submit a private GitHub repository containing:
 - main.py and all required LCD helper files
 - Wiring diagram or clear wiring photo
 - Wi-Fi and webserver setup instructions
 - Instructions for using every web page control
 - Screenshots of the completed web page and LCD
 - Demonstration video

6. Submission and Academic Integrity

The demonstration video must show:

- Live temperature, humidity, and distance readings
- Temperature and distance displayed on the LCD through web buttons
- Servo angle controlled using the web slider
- Custom text sent from the browser to the LCD